TownPay

Pay. Sell. Stay in sync.

ITMDA Group Project - Documentation

Project screenshots | GitHub repository

October 2026

About the application

TownPay is a mobile payments and small-business prototype built with React Native (Expo) and React Navigation. It lets a merchant sign in with a PIN-protected account, send money, pay and receive through QR codes, manage product inventory, ring up sales, and review a daily sales summary. Payments are simulated and no real money moves, but each feature works end to end inside the app and shares one transaction store, so a QR payment or a till sale appears on the Home dashboard.

Quick link

GitHub repository https://github.com/demitrist/TownshipPay-App

Full details are in section 2 at the end of this document.

Demo login for testing

Email	**demo@townpay.app**
Password	**townpay**
PIN	**1234**

Any valid email + 6-character password also signs in.

Team and feature ownership

Demitri	Navigation, auth, theme, profile
Christopher	Home, send money, transactions
Kateho	QR payments and receipts
SG	Inventory, sales, daily summary

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1. Project Screenshots

The screenshots below were taken from the running TownPay application and follow a typical user journey: signing in, using the dashboard, sending money, paying with QR codes, managing inventory, and completing a sale.

1.1 Launch, registration and sign-in

The app opens on a branded splash screen and then the sign-in form. New merchants can create a profile from the registration screen. Registration is a mock sign-up, so the details stay on the device.

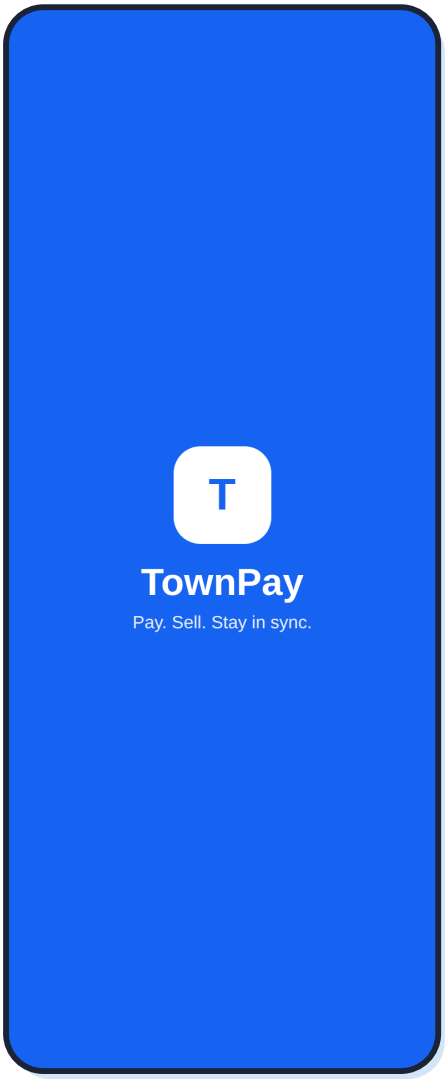


Figure 1: Splash screen
Branded launch screen with the TownPay logo and tagline.

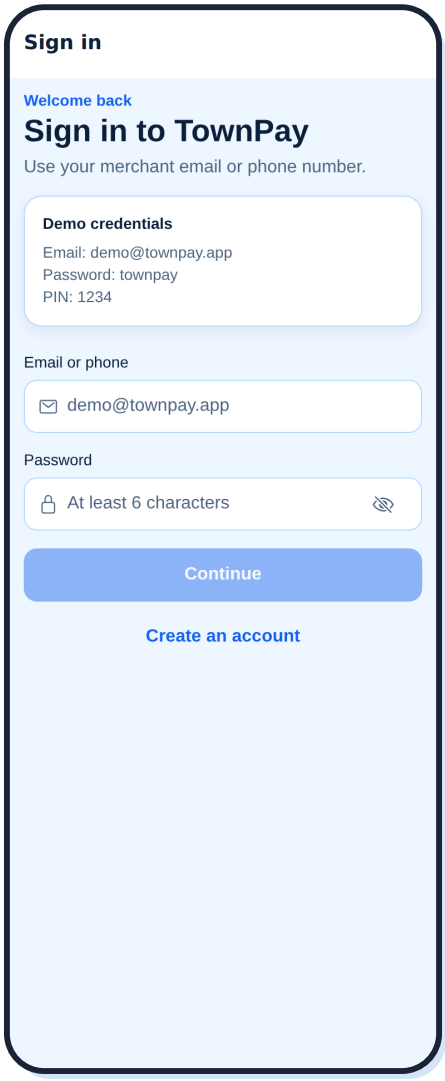


Figure 2: Sign in
Email or phone with password, plus a demo-credentials card for testers.

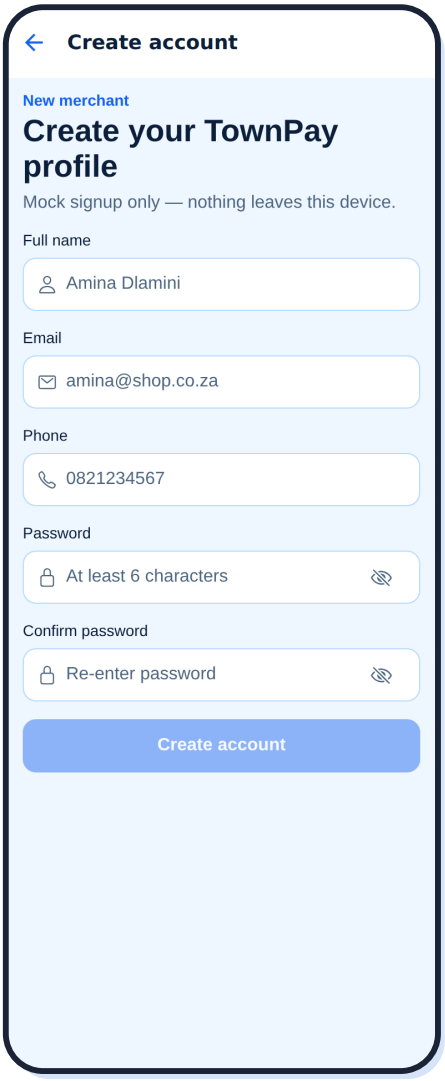


Figure 3: Create account
Full name, email, phone, password and confirmation for new merchants.

What this demonstrates

- Branded splash screen followed by a sign-in form with an email/phone and password check.
- Registration form with validation for name, email, phone and matching passwords.

1.2 PIN security and the home dashboard

After signing in, the user confirms a 4-digit PIN on a custom keypad. The Home dashboard then shows the wallet balance, quick actions, connection status with a sync control, and the latest transactions.

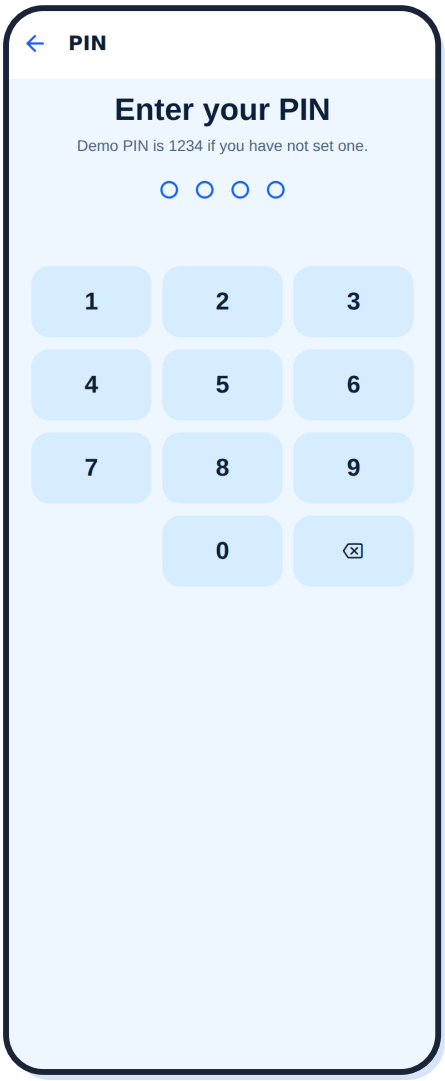


Figure 4: PIN entry
Four-digit PIN keypad that protects access to the wallet.

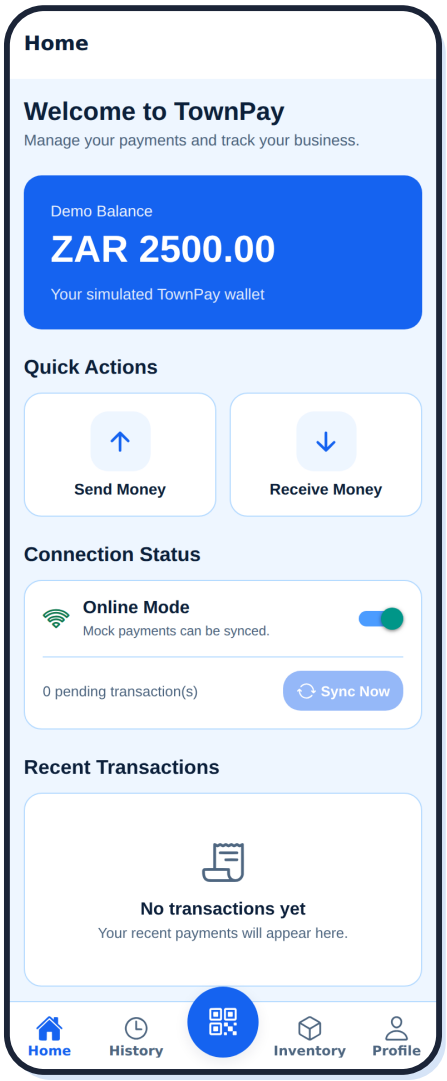


Figure 5: Home dashboard
Demo balance, Send and Receive quick actions, online/offline toggle and Sync Now.

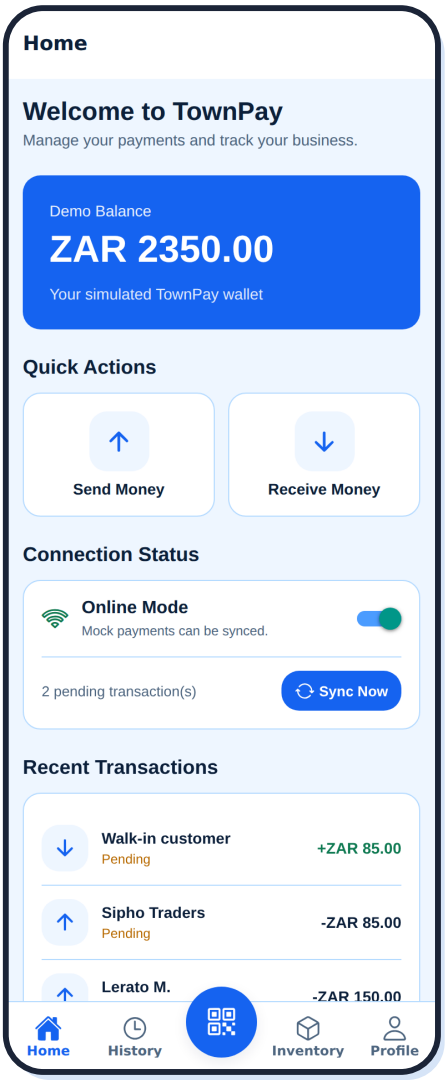


Figure 6: Dashboard after activity
Recent transactions with Pending or Synced status, and the pending count.

What this demonstrates

- A PIN keypad adds a second security step after sign-in.
- The dashboard shows balance, quick actions, online/offline mode and pending-transaction sync.
- Recent transactions carry Pending or Synced status badges.

1.3 Send money

Users send a simulated payment by entering a recipient, an amount and an optional note. The form validates the input, deducts the amount from the demo wallet and records the payment in the transaction list.

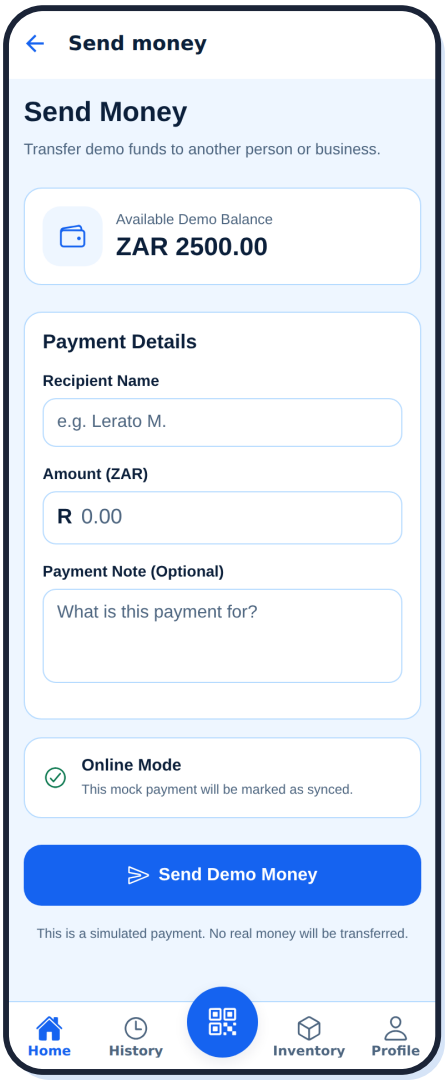


Figure 7: Send money form
Shows the available balance and the payment detail fields.

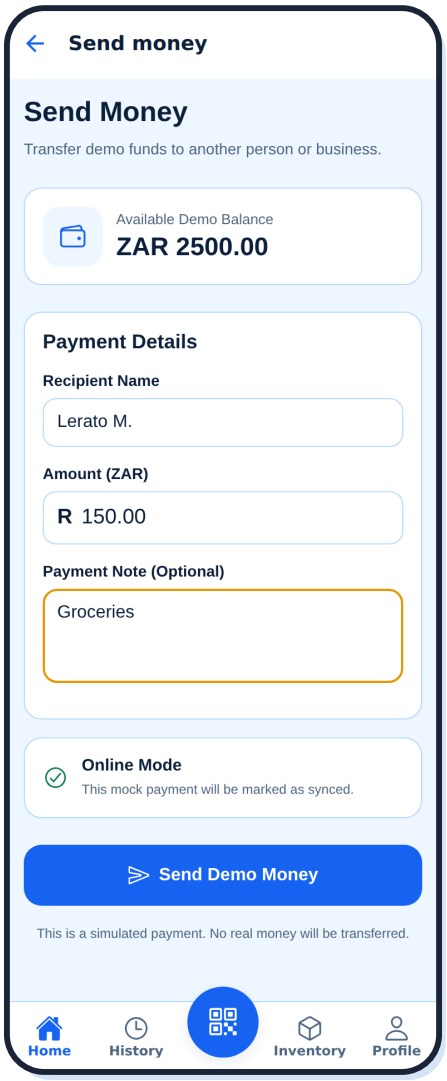


Figure 8: Form completed
Recipient Lerato M., amount R 150.00 and the note "Groceries".

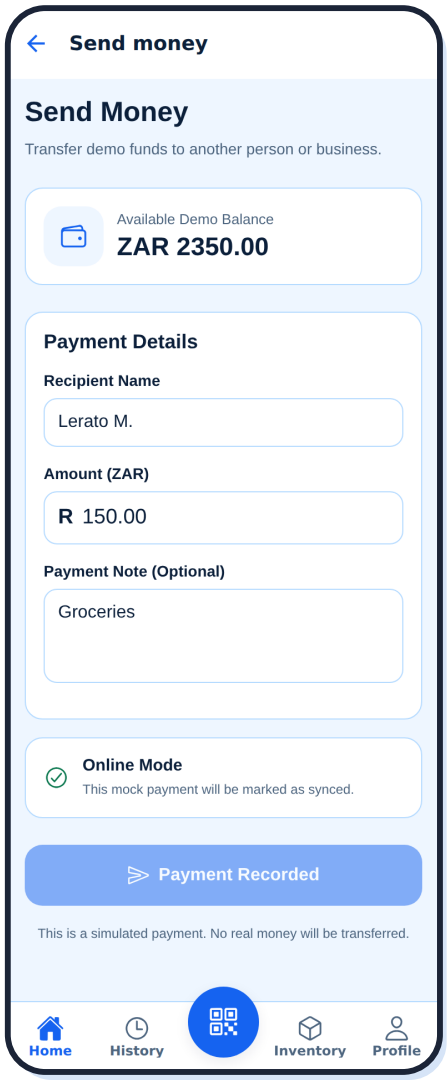


Figure 9: Payment recorded
Balance updated to ZAR 2350.00; online mode marks the payment as synced.

What this demonstrates

- Form validation on recipient and amount before a payment can be sent.
- The wallet balance updates immediately after sending.
- The payment is recorded and marked synced because the app is in online mode.

1.4 QR payments: My QR, scan and confirm

The QR tab, the centre button on the tab bar, gives each merchant a personal QR code to receive money. A scan leads to a confirmation step that shows the payee and amount before anything is charged.

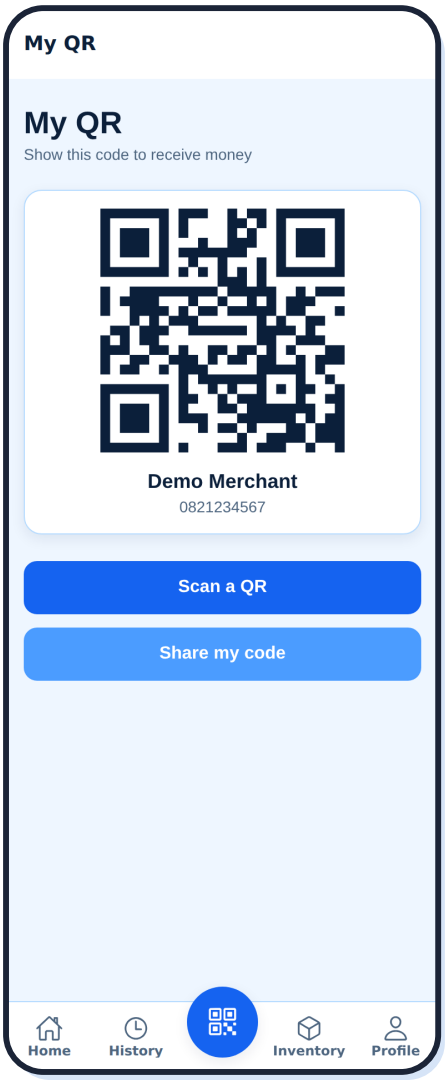


Figure 10: My QR
Personal QR code with the merchant name and number, plus Scan and Share buttons.

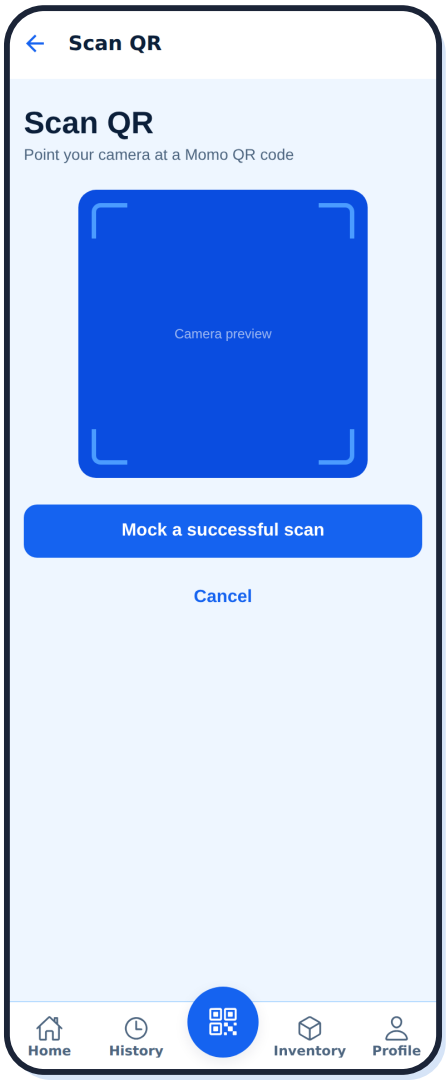


Figure 11: Scan QR
Scanner frame with a mock-scan button so the flow can be demonstrated anywhere.

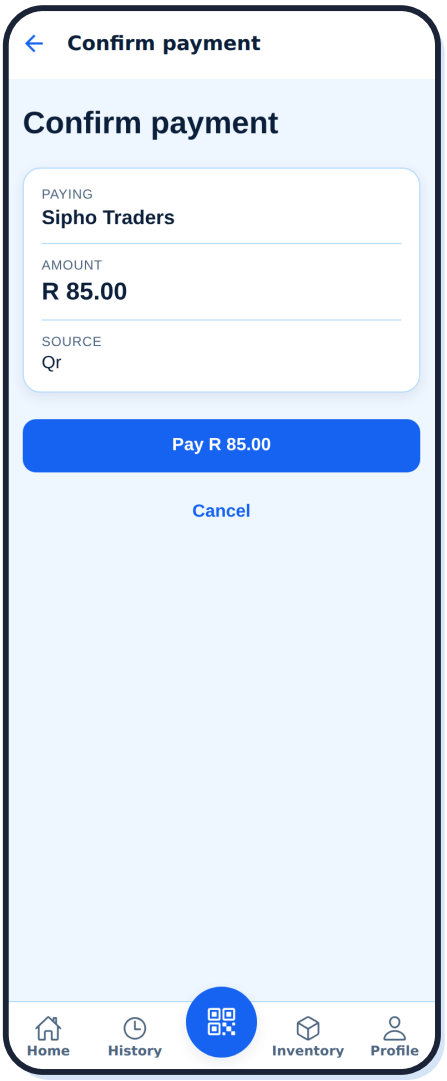


Figure 12: Confirm payment
Paying Sipho Traders R 85.00 via QR, with Pay and Cancel options.

What this demonstrates

- Personal QR code generated for the signed-in merchant.
- Scan flow leads to a confirmation step showing payee and amount before paying.
- A mock-scan button lets the flow be demonstrated without a camera.

1.5 Payment confirmation and receipt

Confirming a QR payment shows a success screen with a status badge and a unique reference, and a receipt that can be viewed or shared.

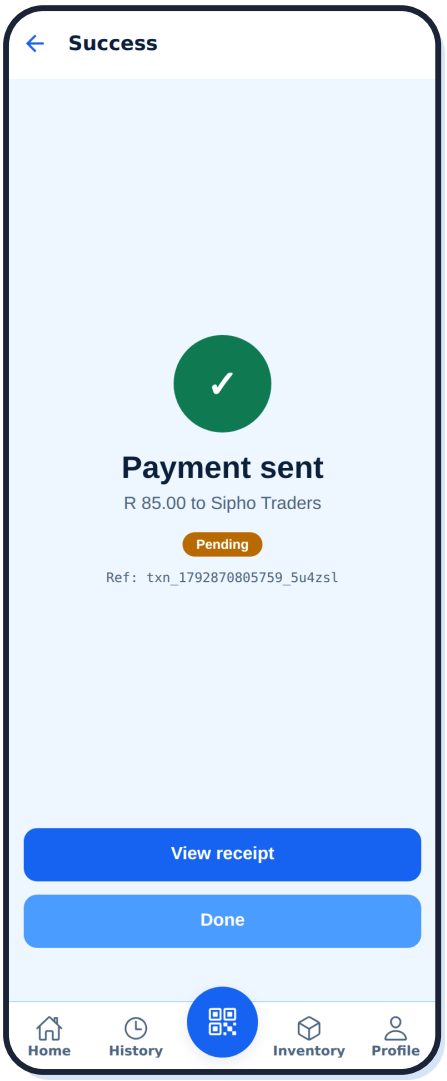


Figure 13: Payment sent
Success state with a Pending badge and the transaction reference.

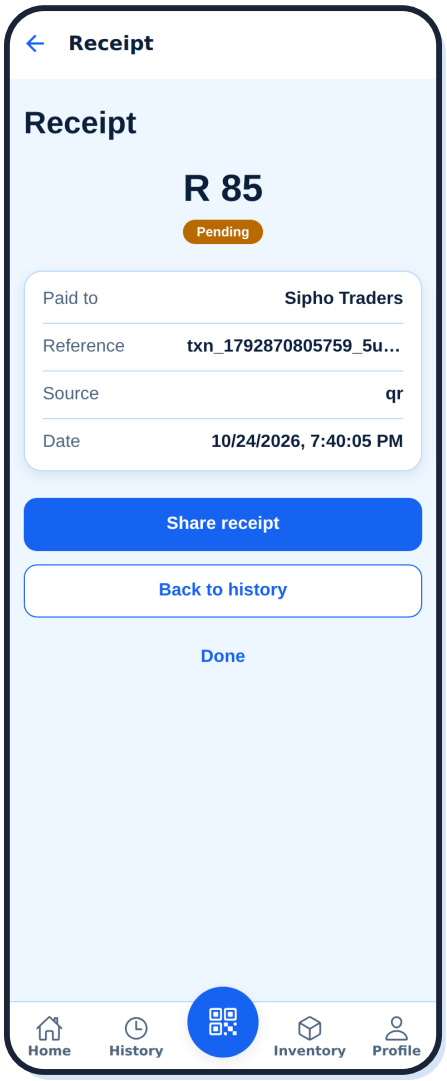


Figure 14: Receipt
Payee, reference, source and date, with Share receipt and Back to history.

Shared transaction store

Every confirmed QR payment is saved as a pending transaction. It then appears in Recent Transactions on the Home dashboard (Figure 6) together with sales and sent payments, and can be marked as synced with the Sync Now control.

What this demonstrates

- Success screen with a unique transaction reference and a Pending status.
- Receipt screen with payee, reference, source and date, and a Share receipt option.

1.6 Inventory: product list and adding an item

The Inventory tab is the hub for small-business tools. It lists every product with its SKU, price and stock level, warns about low or out-of-stock items, and links to adding items, making a sale and the sales summary.

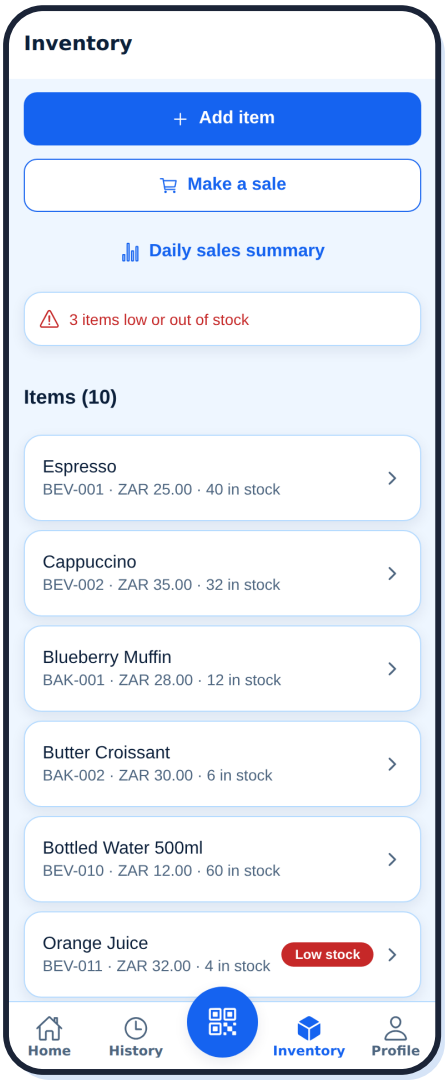


Figure 15: Inventory list

Ten starter products, quick actions and a low-stock warning banner.

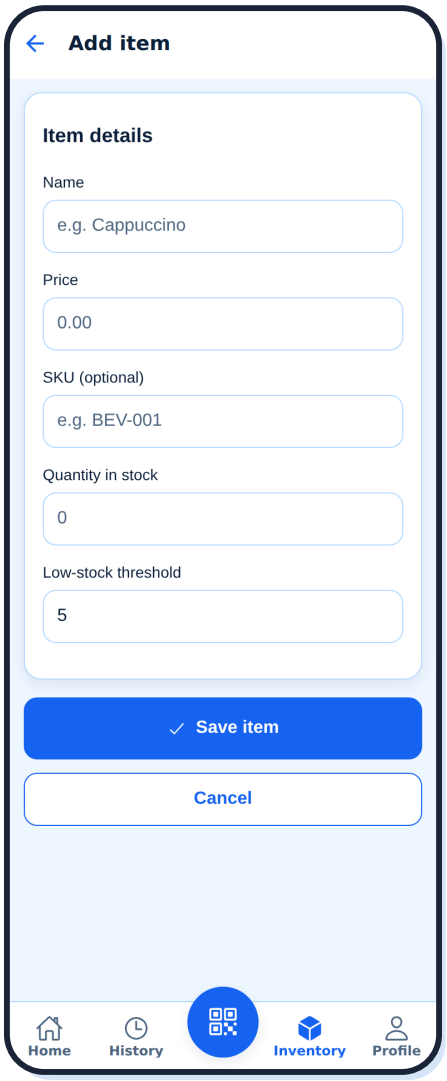


Figure 16: Add item form

Name, price, SKU, quantity and low-stock threshold fields.

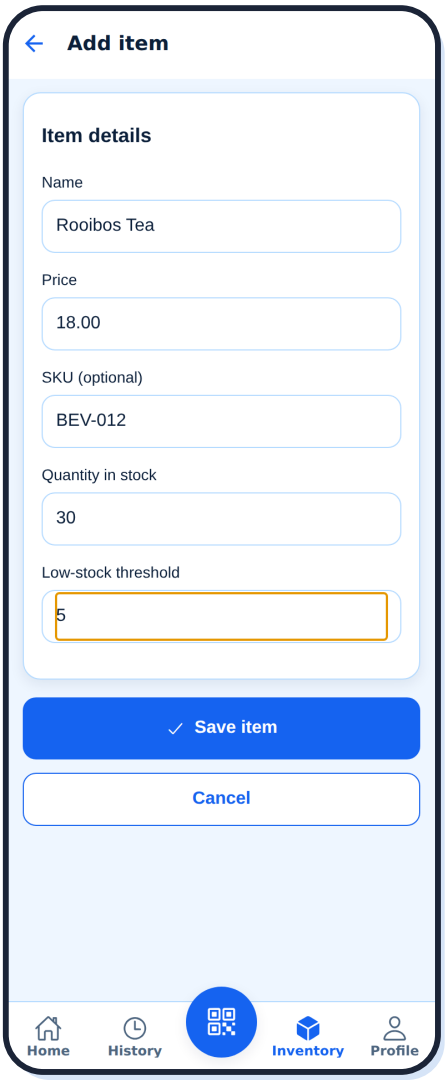


Figure 17: Item details entered

A new product, Rooibos Tea, ready to save.

What this demonstrates

- Product list with SKU, price and stock level for each item.
- Low-stock banner and badges warn when items run low or run out.
- Add-item form with field validation for price and quantity.

1.7 Updated stock, editing items and starting a sale

Saved items appear in the list immediately. Selecting any product opens an edit form that is pre-filled with its current details. The Make a sale screen lets the user build a cart by tapping products.

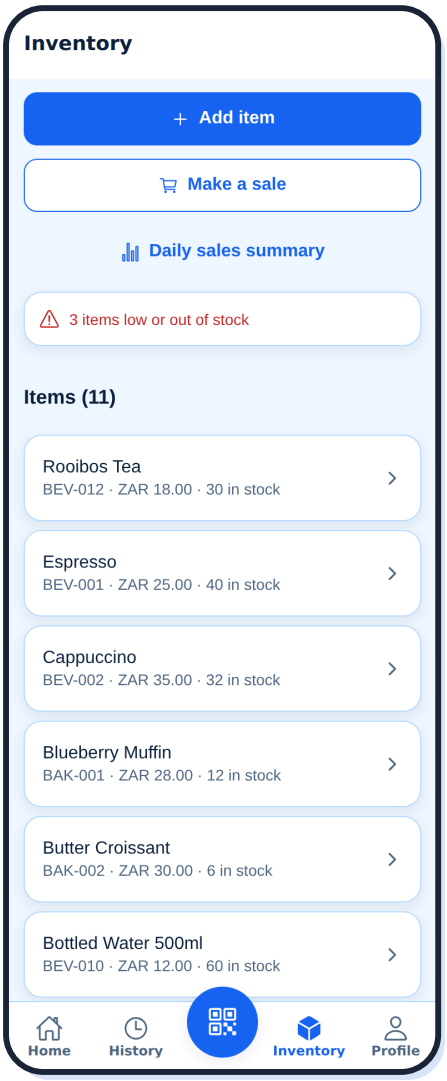


Figure 18: Item added

The list now shows 11 items, with Rooibos Tea at the top.

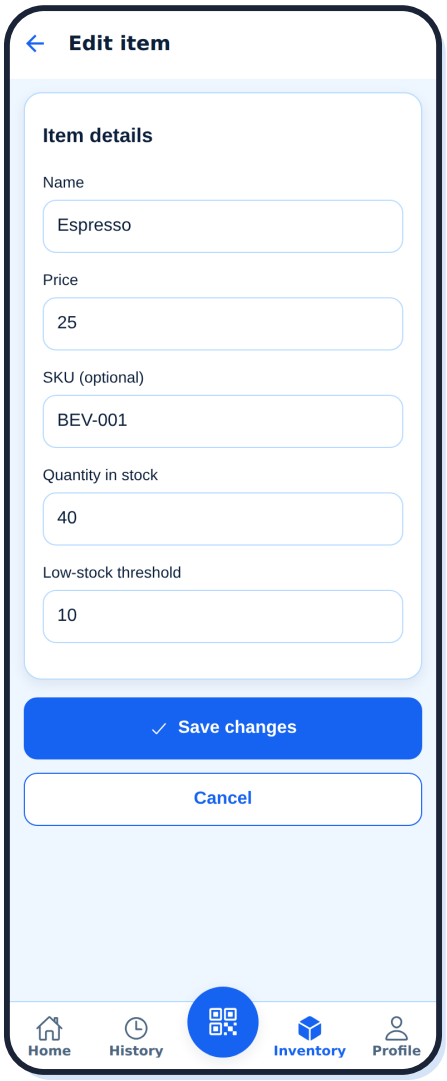


Figure 19: Edit item

Espresso loaded into the form so price, stock and threshold can be changed.

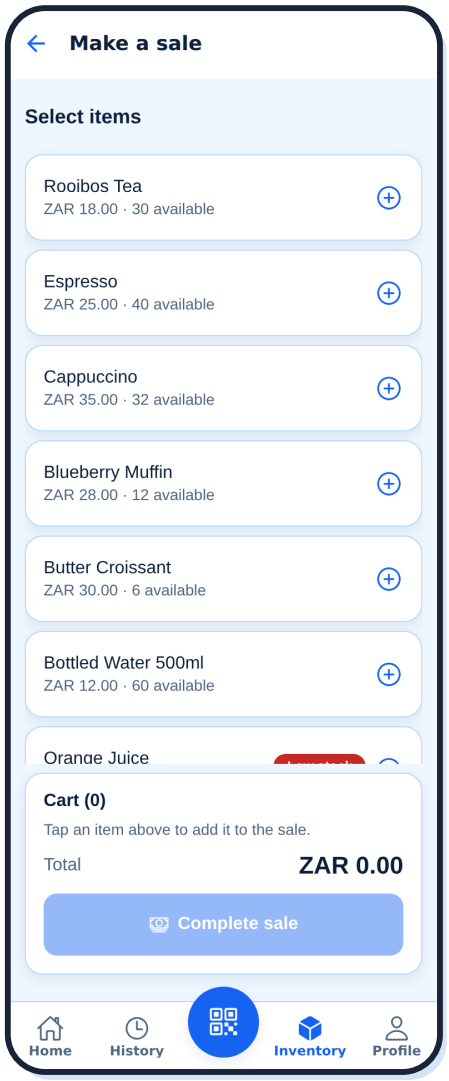


Figure 20: Make a sale

Product picker with live availability, low-stock badges and an empty cart.

What this demonstrates

- New items appear in the product list straight away.
- The edit form is pre-filled from the selected product.
- Make a sale shows live availability and starts with an empty cart.

1.8 Completing a sale and the daily sales summary

Tapping products adds them to the cart with quantity controls and a running total. Completing the sale reduces stock, records a transaction and updates the daily summary, including low-stock alerts.

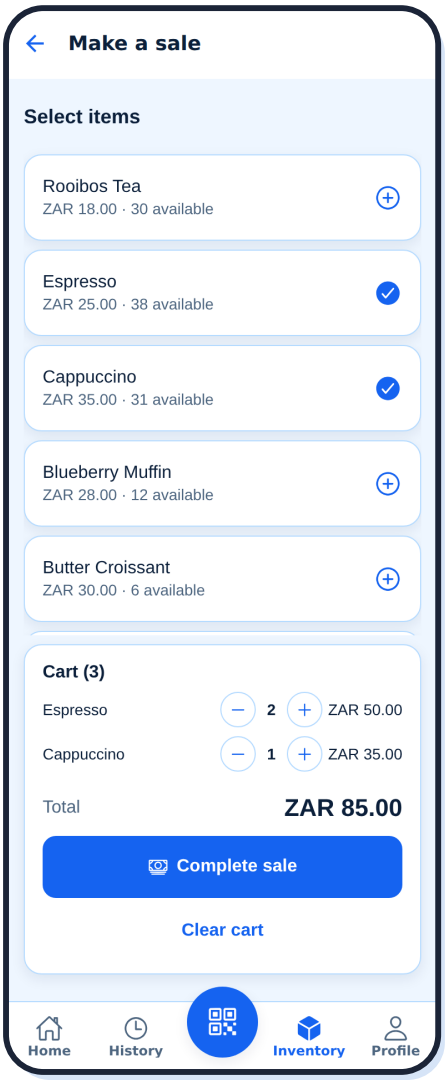


Figure 21: Cart built
Two Espressos and one Cappuccino for a total of ZAR 85.00.

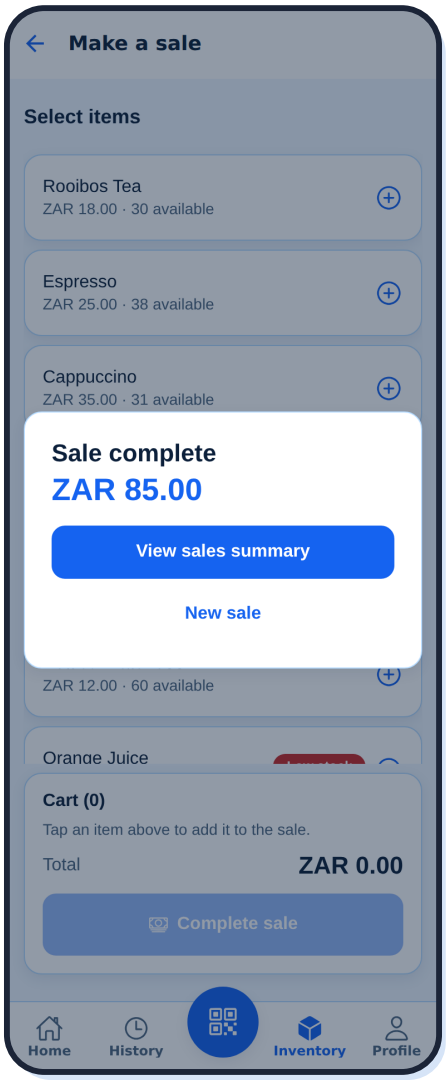


Figure 22: Sale complete
Confirmation dialog with the sale total and a shortcut to the summary.

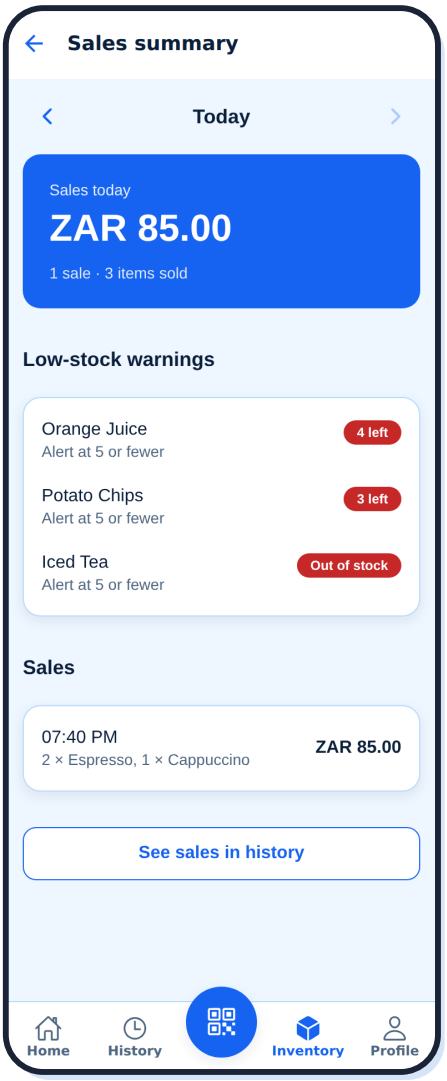


Figure 23: Daily sales summary
Sales today, items sold, low-stock warnings and the sales list.

What this demonstrates

- Cart with quantity controls and a running total.
- Completing a sale reduces stock and records a transaction.
- Daily summary with sales total, items sold, low-stock warnings and the sales list.

1.9 Profile and session management

The Profile tab shows the signed-in merchant and lets them log out, which returns to the sign-in screen.

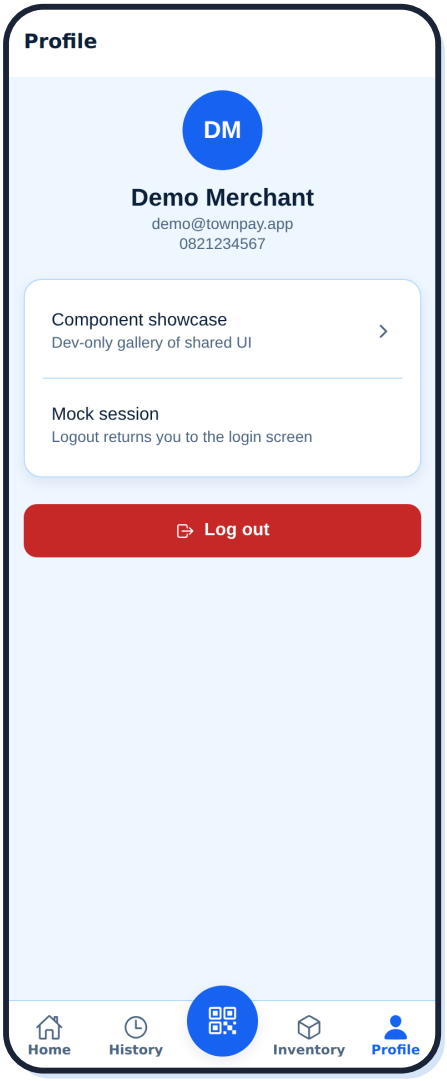


Figure 24: Profile

Merchant details, developer tools and a Log out button.

Features demonstrated

Feature	Figures
Splash, registration, sign-in and PIN	**1 - 4**
Home dashboard, balance and sync status	**5 - 6**
Send money with validation	**7 - 9**
QR code, scan, confirm, success and receipt	**10 - 14**
Inventory list, add item and edit item	**15 - 19**
Make a sale and daily sales summary	**20 - 23**
Profile and log out	**24**

Technology

React Native with Expo SDK 58, React Navigation (native stack and bottom tabs), React Context for authentication, transactions and inventory, AsyncStorage for the mock session, and Ionicons.

2. GitHub Repository Link

Repository URL (click to open)

https://github.com/demitrist/TownshipPay-App

Repository name	**TownshipPay-App**
Default branch	**main**

Running the project locally

1. Clone the repository and open the project folder.
2. Run `npm install` to install dependencies.
3. Run `npx expo start` and open the app in an emulator or in Expo Go.
4. Sign in with the demo credentials shown on the cover page.